



P13

The Wall of Scaling Is a Paradigm Wall, Not a Physical Limit

Scaling 的墙是范式墙：四面墙与五条破墙路径

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The Wall of Scaling Is a Paradigm Wall, Not a Physical Limit

Working Title (EN): The Wall of Scaling Is a Paradigm Wall, Not a Physical Limit: Four Bottlenecks, Five Breakout Paths, and a Three-Section Phase Transition

Topical paper P13 • a direct extension of flagship paper P0b Structure + Scale Is the Intelligence • UDOS Reasoning Engine v7.5.0 • 2026-09-20

In one sentence: the much-discussed "scaling wall" hits not the physical limit of intelligence but the **carrying limit of a fixed paradigm (Transformer + dense compute + internet text)**. Breaking through is not knocking the wall down but switching to a different growth curve—**changing structure, not stopping the stacking**.

Evidence scope (stated once for the whole paper): the numbers the UDOS engine produces itself all come from reports7/*.json, ordinary CPU, fixed random seed, reproducible, marked at three levels: verified (guarded by automated tests), cpu-proto (CPU-prototype trend, constants not calibrated across multiple seeds), unverified (external reports / one-sided claims in papers, used only as motivation). Single-seed results are always written as "pilots," and ≥ 30 seeds with 95% confidence intervals are required before submission. **External literature and industry numbers were verified online one by one on 2026-09-20;** any number appearing only in a press release or a paper's self-report without independent reproduction is explicitly marked "reported by the paper / disclosed by the company, not independently verified."

Abstract

"Has the Scaling Law hit a wall?" has become the most divisive question in the large-model field in 2026. This paper does not directly take a side but first decomposes the "wall" into four **separately measurable** dimensions: the data wall (limited high-quality corpora, entropy saturation of synthetic data), the compute/energy wall (dense matrix multiplication approaching thermodynamic and grid constraints), the architecture wall (the nearly decade-old structures such as attention and residuals becoming bottlenecks), and the returns wall (within the same architecture, marginal capability gains decrease for every order-of-magnitude increase in scale). We argue these four walls share one premise—structure is frozen. What they measure is the phenomenon of "walking one curve to its end within a fixed paradigm," so they are **paradigm walls**, not physical limits of intelligence.

Under the three-section framework of the companion paper P0b, the wall corresponds precisely to the second section of the curve—the structural phase-change inflection: in the first section the structure is right and scale small, capability growing along a power law; in the second the old structure cannot hold and the organization must change; in the third growth continues on the new curve until the next wall is hit. Breaking through is therefore "switching curves" rather than "running faster on the old curve." We converge the 2025–2026 breakout attempts into five paths and give falsifiable hypotheses: (P0) data efficiency, raising effective experience through collection structure rather than collection budget—in UDOS controlled pilots, Coverage-aware active collection at the same budget raised yield from 0.667 to 0.854 and experience density from 7.80 to 11.38 (+46%, verified pilot); (P0) architecture reconstruction, including depth-dimension attention (Attention Residuals, paper reports about $1.25\times$ compute efficiency, arXiv:2603.15031, not independently verified), nonlinear attention expansion (Nexusformer, arXiv:2604.19147), and linear-softmax hybrids (InclusionAI Ling 2.5); (P1) the training paradigm, shifting from pretraining scaling to chain-of-thought/test-time scaling and replay-based self-improvement

(Dream-RL, arXiv:2609.14858, paper reports up to $162\times$ fewer agent calls); (P1) system-level scaling and agent governance (UDOS P1/P2/P9/P10/P11); (P2) the hardware paradigm, the time-scaling route represented by Huawei's "Tau (τ) Law" (He Tingbo et al., ChinaXiv, 2026-05).

We also did one necessary fact check: the widely circulated claim that "compute increased about $10\times$ from GPT-4 to GPT-5" lacks public evidence; third-party Epoch AI-scoped estimates put it at about $2\text{--}3\times$ (GPT-5 total training compute about $5e25$ FLOP vs GPT-4 about $2e25$ FLOP, unverified), and "fleet total GPU count" is often conflated with "single-training compute." This paper does not adopt the $10\times$ figure. Finally we give five predictions that can kill this paper's conclusions and four counter-arguments.

Keywords: scaling law; data wall; architecture wall; phase change; attention residuals; test-time scaling; RSI; evidence grading; falsifiable prediction

Structured abstract (background problem $\rightarrow$ method $\rightarrow$ evidence/results $\rightarrow$ contributions)

- **Background problem:** diminishing marginal returns under the same architecture are read as "Scaling is dead," but that conclusion mistakes "the limit of a fixed paradigm" for "the limit of intelligence."
 - **Method:** operationalize the wall into four measurable dimensions; explain uniformly with P0b's three-section curve (power law $\rightarrow$ phase change $\rightarrow$ saturation/negative returns); converge breakout attempts into five paths, each with a pre-registered falsifiable hypothesis; verify external numbers one by one.
 - **Evidence/results:** UDOS's own evidence is controlled CPU pilots (yield $0.667\rightarrow 0.854$, density $+46\%$, on the wrong structure 8-agent throughput only $0.695\times$, equal-weight bad-expert error differs about $23\times$); external evidence is split into three tiers: "verified to exist / paper reports not reproduced / no evidence found."
 - **Contributions:** (1) the four-dimensional operational definition of the paradigm wall; (2) the priorities and criteria of the five paths; (3) the testable statement "wall = phase-change point"; (4) a public correction of the $10\times$ compute claim.
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1 Introduction: the metaphor of the wall must first be measured

Kaplan et al.'s 2020 Scaling Laws gave an extremely useful engineering belief: as long as parameters $\$N$, data $\$D$, and compute $\$C$ are scaled together, loss falls along a predictable power law [CITATION NEEDED: Kaplan 2020, arXiv:2001.08361, version-volume master reference library]. Chinchilla then added the constraint that "data and parameters must grow in proportion" [CITATION NEEDED: Hoffmann et al. 2022, arXiv:2203.15556]. By 2024–2026, three phenomena made "hitting the wall" a popular narrative: high-quality internet text approaching exhaustion, the power and cluster scale of frontier-model training approaching real constraints, and narrowing intergenerational gaps among same-generation architectures.

Metaphor cannot replace measurement. If the word "wall" is to enter engineering discussion, it must answer three things: on which variable the wall sits, how its height is measured, and what is on its other side. This paper's stance is:

What the four walls measure is the same thing—diminishing marginal returns along a single scale axis when structure is frozen. They are paradigm walls, not physical-limit walls.

This stance directly inherits P0b's central formula:

$$\text{Capability} \approx F(S; N, D, C), \quad S = (T, G, M) \tag{1}$$

where SS is the structure triple (connectivity topology ST , geometry/spatial organization SG , modular boundaries SM), and N, D, C are nodes/parameters, data/experience, and compute/energy respectively. Fixing SS and only enlarging N, D, C moves the system along one established curve; changing SS switches to another curve. All the "breakout paths" this paper discusses essentially answer "which structural variable to change."

1.1 What this paper does not claim

To avoid being read as yet another "long live Scaling" defense, three boundaries are drawn first:

1. This paper **does not deny the reality of the walls**. The four constraints—data, energy, architecture, returns—all have observable evidence; denying them equals denying engineering reality.
2. This paper **does not predict infinite growth beyond the wall**. Changing structure only pushes the wall back, and the new structure will hit the next wall; the three-section curve is open and repeatable, not a one-way road to a singularity.
3. This paper **does not treat external vendors' efficiency multiples as established facts**. Any $1.25\times$, $162\times$, $3.2\times$ not independently reproduced is always marked "reported by the paper / disclosed by the company."

2 Related work and lineage

Scaling and data constraints. Kaplan-style power laws and Chinchilla's compute-optimal ratio constitute the "Old Testament" [CITATION NEEDED]. Two later threads directly relate to this paper: on the data side, "model collapse" research on duplicated and synthetic data points to entropy saturation in self-consuming data [CITATION NEEDED: Shumailov et al., Nature 2024]; on the architecture side, sparsification (Switch Transformer, arXiv:2101.03961) proves "large total parameters, small activated parameters" is feasible [CITATION NEEDED].

Structural priors are performance. BPE (arXiv:1508.07909) and SentencePiece (arXiv:1808.06226) are input structural priors; attention (arXiv:1706.03762) is a sequence structural prior; MoE routing is a capacity structural prior. A batch of 2026 works began directly rewriting these priors: Attention Residuals rewrites the depth-dimension residual aggregation (arXiv:2603.15031, verified to exist); Nexusformer replaces linear Q/K/V projections with nonlinear Nexus-Rank (arXiv:2604.19147, verified to exist); JTok proposes token-indexed parameters as an orthogonal scaling axis decoupled from FLOPs (arXiv:2602.00800, verified to exist); SCONE gives embedding-layer/n-gram expansion strategies (arXiv:2502.01637, verified to exist).

Test-time and self-improvement scaling. Chain-of-Thought (arXiv:2201.11903) opened reasoning-chain elicitation [CITATION NEEDED]; 2026's Dream-RSI treats "discovery-tree replay" as a zero-cost simulator to improve exploration policy (arXiv:2609.14858, verified to exist, exact title Recursive Self-Improvement through Evolving Worlds). The RSI lineage and the seven major vendors' routes are the topic of companion paper P14, and this paper cites its conclusions only in path three.

Hardware time scaling. Huawei proposed the "Tau (τ) Law" at IEEE ISCAS on 2026-05-25; the paper A Time Scaling Theory for Multi-Layer Electronic Systems signed by He Tingbo was submitted to ChinaXiv, arguing for replacing geometric scaling with scaling of the system-level time constant τ , disclosing 381 chips designed and mass-produced from 2020-05 to 2026-05, and predicting that by 2031 high-end chip transistor density (155→400+ MT/mm²) will reach the equivalent of a 1.4nm process (company/conference disclosure, multi-source reports verified, specific predicted values unverified).

3 The four walls: decomposing the metaphor into four measurables

Figure 1 The four walls of scaling

Figure 1 The four walls are not four independent walls but four projections of the same "paradigm wall." The measurement scope of each is defined below.

3.1 The data wall: effective tokens, not total tokens

The strict statement of the data wall is not "internet text is finite" but **the growth rate of the effective token stream $D_{\{\mathrm{eff}\}}(t)$ after de-duplication, de-contamination, and de-debasement is finite:**

$$D_{\{\mathrm{eff}\}} = \sum_i \mathbb{1}[\text{novel}_i] \cdot q_i, \quad q_i \in [0,1] \tag{2}$$

where novel_i indicates whether the i -th datum brings new state coverage relative to the existing data pool, and q_i is the quality score. When newly added data are mainly replicas of old scenes, $\text{novel}_i \rightarrow 0$: total hours rise while $D_{\{\mathrm{eff}\}}$ does not. Synthetic data stack an entropy-saturation risk on top: retraining a model on model outputs shrinks distribution variance generation by generation. The data wall is therefore first a **collection-structure** problem, not a collection-budget problem—precisely the entry point of path one.

3.2 The compute/energy wall: linear loss reduction needs exponential energy

Under the power-law loss $L(C) \approx L_{\infty} + (C/C_0)^{-\alpha}$, the compute needed to press loss down by another fixed amount ΔL grows on the order of $1/\Delta L^{1/\alpha}$; while data-center power supply, cooling, and grid access are linear resources. The wall's height can be measured by the "marginal joules per unit capability increment" $dE/d(\text{capability})$. This wall is a real physical constraint, but it constrains the specific structure of **dense fully activated compute**; sparse activation, on-demand test-time compute, and specialized hardware all change the $dE/d(\text{capability})$ curve itself.

3.3 The architecture wall: the decade-old standard becomes a bottleneck in the grammar of growth

The architecture wall means: the connection methods defaulted to as a "neutral base"—the quadratic complexity of softmax self-attention, the indiscriminate accumulation of residual connections across layers, the dense FFN—introduce structural discounts as depth/sequence length grows. The Attention Residuals paper states the latter as "PreNorm dilution": hidden-state magnitude grows linearly with depth, and a useful representation at layer 5 is diluted by the other 45 layers by layer 50 (arXiv: 2603.15031 abstract and secondary interpretations, paper verified to exist; mechanism description per its abstract). The architecture wall is the most easily overlooked because it disguises itself as "not enough compute."

3.4 The returns wall: diminishing marginal gains within the same architecture

The returns wall is the aggregate manifestation of the first three walls on the capability axis: with fixed \$\$\$, $\partial \text{capability} / \partial \log N$ falls monotonically with $\log N$. P0b gave a sharp micro-example on the CPU prototype: on CPU-intensive tasks, concurrent agents from 1→2 throughput $1.594\times$, from 2→8 not rising but falling to $0.695\times$ (reports7, single-seed pilot); while I/O-intensive tasks over the same interval were approximately linear at $7.99\times$. **The scale axis is exactly the same, and the sign of returns is determined by structural matching**—the returns wall is the shadow of the structure wall.

4 The five breakout paths: actions, evidence, and falsifiable hypotheses

Figure 2 The three-section curve and curve switching

Figure 2 The wall corresponds to the phase-change inflection of the three-section curve. Breaking through = making the system jump to a new curve, rather than continuing to move right on the old curve.

4.1 Path one (P0): data efficiency—change collection structure, not collection budget

Action: upgrade data collection from passive recording to Coverage-aware active scanning—first do a distribution diagnosis of the existing data pool, lock state-coverage blind spots (long-tail objects, flexible manipulation, complex forces), then schedule targeted supplementary collection; meanwhile use automated quality checks to discard invalid fragments.

UDOS controlled evidence (verified pilot, reports7, 48 episodes / 144 minutes, oracle 226 units):

Figure 3 Active-collection yield and density

Figure 3 At the same collection budget, passive collection yield 0.667, experience density 7.80 units/hour; active blind-spot scanning yield 0.854, density 11.38 (+46%); the same-skewed-pool greedy ablation comparison is 0.7917/0.889, showing the gain comes from "blind-spot targeting" rather than "collecting harder." Single-seed pilot, ≥ 30 seeds needed before submission.

External same-direction work (unverified, paper reports): JTok's token-indexed parameters claim to add capacity without extending backbone FLOPs (arXiv:2602.00800, paper verified to exist; the specific "about 1/3 compute saved" figure was not confirmed from the original in this verification, marked [RESULT NEEDED]); SCONE's n-gram embedding expansion provides another embedding-layer scaling axis (arXiv:2502.01637).

Falsifiable hypothesis H1: with fixed collection budget, the yield improvement $\Delta Y > 0$ of active scanning over passive collection is significant at ≥ 30 independent scenario seeds and 95% confidence; if in multi-scenario reproduction the confidence interval of ΔY crosses 0, the core claim of path one is overturned.

4.2 Path two (P0): architecture reconstruction—changing the grammar of growth

Actions and evidence (all external paper reports, not independently verified):

Direction	Work	Reported core breakthrough	Verification status (2026-09-20)
Depth-dimension attention	Attention Residuals (Kimi/Moonshot)	Replace fixed residual accumulation with softmax attention over historical layer outputs; Block AttnRes reduces complexity from $\mathcal{O}(Ld)$ to $\mathcal{O}(Nd)$; paper reports about $1.25\times$ compute efficiency on 1.4T-token pretraining, inference latency increase $<2\%$ (Kimi Linear of 48B total/3B activated)	arXiv:2603.15031 verified to exist; multiples are paper reports
Nonlinear attention expansion	Nexusformer	Three-stage nonlinear Nexus-Rank mapping; zero-initialized blocks support lossless structural growth; reports matching Tokenformer perplexity with at most about 41% (parameter/compute scope)	arXiv:2604.19147 verified to exist; specific percentages are paper reports
Linear-softmax hybrid	InclusionAI Ling 2.5 (1T, 2026-02)	Lightning Attention recurrent linear path + DeepSeek-style MLA, reducing long-range KV cache	Secondary reviews (Raschka et al.) consistent across multiple sources, original report not read
Linear attention ("SeedFold")	Materials call ByteDance SeedFold, cubic $\rightarrow$ quadratic complexity	—	This search found no corresponding paper/product, marked no evidence [CITATION NEEDED], this paper does not accept it

The isomorphic evidence on the UDOS side is P0b's " $23\times$ vs 0.5%": integrating one unqualified expert with equal weight raises MSE from calibrated weighting's 0.0092 to 0.2131 (about $23\times$ worsening), while simply increasing the number of qualified members barely moves the error (reports7, \$k\le 2\$ verified). **The lever of changing connection/weighting structure is far larger than adding node count.**

Falsifiable hypothesis H2: under the same data and same training-compute budget, the per-compute perplexity/task-return advantage of the new structure over the old should be independently reproducible by third parties at $\geq 1\text{B}$ parameter scale; if only the original author team can reproduce it, or the advantage disappears after switching data domains, then "changing structure changes the curve" does not hold in that direction.

4.3 Path three (P1): the training paradigm—from pretraining scaling to chain-of-thought/test-time scaling

Pretraining scaling is "having seen more," test-time and replay scaling is "reasoning deeper": one real experience replayed through a world model generates 1000 virtual simulations and tens of thousands of internal thought trajectories. The key mechanism of Dream-RL (Google/DeepMind et al., arXiv: 2609.14858, submitted 2026-09-14) is treating the completed discovery tree as "a replay simulator over an already realized search space," evaluating and improving exploration policy offline on the tree, with the underlying coding agent unchanged; the paper reports up to $162\times$ fewer agent calls than SimpleTES and about $1.7\times$ over the fixed-exploration baseline, and observes that human guidance is instead worse (the above multiples are all paper reports, not independently verified; both authors and third parties stress it is **not** complete RSI where the model modifies its own weights).

Falsifiable hypothesis H3: the returns of replay-based policy improvement should be monotonically non-decreasing with discovery-tree size and show no significant negative transfer when migrated to a new task family; if "dreaming" works only on the original task, it is within-task caching, not a new scaling axis.

4.4 Path four (P1): system-level scaling and agent governance—from single model to organization

When the single-point model's curve flattens, organizing multiple agents into a system opens a new scale axis, but this axis likewise obeys "structure determines shape." The controlled results of six UDOS topical papers (numbers all trace reports⁷, property/contract/unit-test guards verified, extrapolations cpu-proto):

- **Topology (P1):** at million nodes the star center must process about 20.97 million messages with a critical path of about 8.39 million rounds; the layered hybrid topology keeps center load constantly 9 with only 14 rounds—the growth exponent changes from 1 to 0.
- **Stop decisions (P2):** the BFT-lite committee $n \geq 3f+1$, quorum $2f+1$, at $n=7, f=2$ correctly accepts 24 work orders 24/24; 10 property tests verified.
- **Quality saturation (P9):** $mse(k)=a+b/k$, homogeneous expansion beyond the true independent-source count makes the dividend go to zero ($k \leq 2$ verified, $k > 2$ cpu-proto).
- **Handoff contract (P10):** the TransferBundle six fields (Goal/Context/Done/ToDo/Trace/Owner) mechanically validated, 26 contract tests verified; failure-rate field is [RESULT NEEDED].
- **Settlement conservation (P11):** $\text{balance_sum} = \text{paid} - \text{slashed}$ guaranteed by code structure, mixed-task budget 24/paid 16/slashed 1 conservation true, 10 unit tests verified.

Falsifiable hypothesis H4: the closure-rate improvement brought by governance mechanisms (unique Owner, Trace, explicit stop conditions, conservation invariants) should hold in stress tests over agent counts 10^1 to 10^4 ; if beyond some scale threshold the Trace-break rate rises out of control, then system-level scaling likewise has its own wall and the mechanisms this paper gives are insufficient.

4.5 Path five (P2): the hardware and energy paradigm—bypassing the geometric-scaling wall

Huawei's "Tau (τ) Law" argues for no longer betting all performance growth on transistor geometric size, but compressing the signal-propagation time constant τ through full-stack device-circuit-chip-software coordination (logic folding, critical-path shortening, parasitic RC reduction); its paper and launch disclose 381 chips over 6 years and 2031 density equivalent to 1.4nm ($155 \rightarrow 400+$ MT/mm²). This is a typical "change structure, change curve" idea, but all 2031 numbers are **predicted targets, not measurements**, and this paper lists them as unverified. The architecture-level companion is heterogeneous mixing: Transformer handles logical reasoning, linear attention/Mamba-class handles long time series, and the world model handles physical-state prediction.

Falsifiable hypothesis H5: the time-scaling route should, without relying on the most advanced lithography nodes, keep delivering intergenerational improvements on public benchmarks (e.g. token throughput per watt); if mass-produced chips in 2027–2028 cannot reproduce the predicted curve, the τ route is falsified as engineering narrative.

4.6 Priorities of the five paths

Priority	Path	Core action	Evidence maturity
P0	Architecture reconstruction	Change the underlying connection method	External paper reports active, UDOS side has structural-lever micro-verification
P0	Data efficiency	Change collection structure	UDOS has controlled pilots (single seed), external same-direction papers exist
P1	Training paradigm	Test-time/replay scaling	External paper reports, mechanism clear, multiples pending verification
P1	Agent governance	Topology + committee + contract + conservation	UDOS multiple verified tests, scale extrapolation pending
P2	Hardware paradigm	Time scaling + heterogeneous mixing	Vendor roadmaps and predictions, longest realization cycle

The priority ordering principle is the **lever/evidence-maturity ratio**, not technical flashiness.

5 One necessary fact check: "GPT-4→GPT-5 10× compute" does not hold

Figure 4 Compute-multiple check

Figure 4 Comparison of public estimates and rumors. Third-party Epoch AI scope (compiled via TensorFlow 2026-05) estimates GPT-5 total training compute about 5e25 FLOP, GPT-4 about 2e25 FLOP, about 2.5×; and GPT-4.5 by the same scope may exceed GPT-5. All the above are unverified third-party estimates.

"Compute increased about 10× but performance not proportionally" is one of the most cited numbers in the four-walls narrative. This verification (2026-09-20) found no public scope supporting 10× and instead found three common scope confusions:

1. **Fleet size ≠ single-training compute.** "OpenAI about 200,000 GPUs" / "a single 100,000-card training task" is infrastructure capacity or orchestration scale and cannot be directly taken as a generation's pretraining compute.
2. **Pretraining compute ≠ total training compute.** RL/post-training share ranges in different estimates from 10% to 200%, and different scopes can make the multiple differ by more than half.
3. **Input compute ≠ effective compute.** Utilization (MFU), floating-point precision, and MoE activation rate all change "the capability bought by the same FLOPs."

Therefore this paper **does not use 10×** in any chart, switching to "about 2–3× (third-party estimate, unverified)" with the scope explicitly marked. This correction is itself a methodological stance: before discussing the "wall," first ensure the ruler measuring the wall is accurate.

6 Discussion: the wall is a phase-change point, breaking through is switching curves

P0b's three-section curve gives the four walls one unified position: they all occur in the second section. The brain uses cortical folding for the wiring wall, chips use multiple cores for the power wall, models use MoE routing for the parameter wall, and agent systems use layered topology for the center fan-in

wall—history repeatedly presents the same pattern: **saturation is not an end signal but a "time to change structure" signal.**

This also limits the conditions under which optimistic statements like "Scaling is not dead" hold. Anthropic's judgment about "still aggressively accelerating in 2026" (secondary report, [CITATION NEEDED]) holds only when "acceleration happens on a new curve"; if acceleration means feeding more electricity on the old curve, it is precisely negated by the returns wall. **Scaling Laws are not dead; they changed their way of living: from running farther on one curve to constantly switching to better curves.**

7 Predictions that can kill this paper and counter-arguments

Five hard predictions (pre-registered):

- F1 (corresponding to H1): in ≥ 30 scenario-seed reproductions, the 95% CI of the active-collection yield gain does not cross 0; crossing 0 negates path one.
- F2 (corresponding to H2): the structural advantage of Attention Residuals/Nexusformer-class holds in third-party $\geq 1B$ reproductions; no independent reproduction within two years downgrades the strong claim "the architecture wall can be broken by changing structure" to an individual case.
- F3 (corresponding to H3): if replay-based RSI shows negative transfer on a new task family or returns flatten with tree size, it is not a new scaling axis.
- F4 (corresponding to H4): if agent governance mechanisms in $\$10^4$ -scale stress tests fall below the control group in closure rate, the governance dividend of system-level scaling has an unidentified wall.
- F5 (master): if there exists a class of capability whose improvement under **all known structural transformations** is strictly governed by the same power law with no room for curve switching, then the "paradigm wall" framework is wrong—that would be the true physical-limit wall.

Four counter-arguments:

1. "The wall is the physical limit; changing structure only improves the constant, the exponent cannot be escaped." Partially agreed— τ , energy, and land are ultimately physical constraints; but this paper argues repeated phase changes keep pushing the limit back, the observable near-future wall is still a paradigm wall, and F5 is the deciding experiment.
 2. "Efficiency multiples are all vendor self-reports, not credible." Agreed for the most part, so this paper marks all external multiples as paper reports/unverified, and conclusions do not depend on any single multiple.
 3. "The active-collection gain only picked easy tasks." This is precisely why the same-skewed-pool greedy ablation (0.7917/0.889) was set; but a single seed is insufficient for conclusions, and F1 already requires multiple seeds.
 4. "The five paths are a parallel list, not a theory." Accepted—this paper's theoretical claim is only one (the wall = the phase-change point of fixed structure), and the five paths are that claim's projection on the 2026 technology map; they can become outdated while the claim need not.
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8 Threats to validity and applicability boundaries

- This paper's UDOS numbers come from a CPU prototype, fixed single seed, pilot point estimates; absolute constants cannot be extrapolated to real LLM clusters, and ≥ 30 seeds with 95% CI are needed before submission (about 64/65 places [RESULT NEEDED]).
 - External papers are mostly 2026 preprints, not peer-reviewed; arXiv numbers are verified to exist, but reported multiples are not independently reproduced.
 - "SeedFold" found no evidence in this verification and has been explicitly removed; if the materials contain other same-class unverified names, they should be treated the same way.
 - Huawei's τ law and each vendor's 2028/2031 targets are predictions, not facts.
 - This paper does not discuss consciousness nor claim any system already has a self; the characterization and safety boundaries of RSI are in companion paper P14.
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9 Conclusion

The wall of scaling is real, but it measures the carrying capacity of a frozen paradigm, not the physical limit of intelligence. After decomposing the wall into the four measurable dimensions of data, compute/energy, architecture, and returns, one can see: the common solution to the first three is changing structure (collection structure, connection structure, the time structure of computation), while the fourth (the returns wall) is the aggregate symptom of the first three structural mismatches. P0b's three-section curve gives the wall a precise position—the phase-change inflection; the five paths of 2025–2026 give the engineering map for opening a door around the wall. The highest priority is not more expensive compute but reconstructing the data-collection structure and the model-connection structure. **The essence of breaking through is not knocking the wall down but opening a new door beside it; and the blueprint for opening that door is structure, not scale.**

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9. Scaling Embedding Layers in Language Models (SCONE). arXiv:2502.01637, 2025 (**verified to exist**).

10. Zheng T. et al. Dream-RSI: Recursive Self-Improvement through Evolving Worlds. arXiv:2609.14858, 2026-09-14 (**verified to exist**; $162\times/1.7\times$ /"human guidance worse" are paper reports).
 11. InclusionAI. Ling 2.5 / Ring 2.5 technical materials, 2026-02 (MLA + Lightning Attention; per Raschka's review and InfoQ secondary materials, original not read [CITATION NEEDED]).
 12. He Tingbo et al. A Time Scaling Theory for Multi-Layer Electronic Systems (Tau/ τ Law). ChinaXiv, 2026-05; released at IEEE ISCAS 2026; 381 chips/155 $\rightarrow$ 400+ MT/mm²/2031 equivalent 1.4nm are company disclosure and prediction (multi-source reports verified the event, predicted values unverified).
 13. Epoch AI / TensorFeed compilation. Has frontier AI training-compute growth actually slowed? 2026-05 (GPT-5 $\approx$ 5e25, GPT-4 $\approx$ 2e25 FLOP estimates, **third-party estimates, unverified**).
 14. "SeedFold": public search this time (2026-09-20) found no corresponding paper or product, **no evidence, not accepted** [CITATION NEEDED].
 15. UDOS companion papers: P0b Structure + Scale Is the Intelligence, P1/P2/P9/P10/P11/P12 (this volume), numbers trace `reports7/*.json`.
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Appendix A Reproduction commands and test index

- Data-flywheel pilot: `reports7/coverage_flywheel*.json` (48 episodes/144 minutes, oracle=226); comparison keys `passive / active / greedy_samepool`.
- Topology extrapolation: `reports7/topology_scale*.json` (six target levels 10 $\rightarrow$ 1e6); BFT: `reports7/bft_lite*.json` ($\$n=7, f=2\$, 24$ work orders).
- Quality saturation: `reports7/expert_mixture*.json` ($\$mse(k)=a+b/k\$, \$a=0.009252, b=-4.233\times 10^{-5}\$$).
- Test guards: P2 10 property tests, P10 26 contract tests, P11 10 unit tests (commands and versions in P12's appendix).

Appendix B Evidence ledger (three tiers)

- **verified (test-guarded)**: BFT 24/24, conservation 240/16/1, contract-field validation, topology-count formulas.
- **cpu-proto (CPU-prototype trend, single seed)**: yield 0.667 $\rightarrow$ 0.854, density 7.80 $\rightarrow$ 11.38, throughput hump 244.172/106.485, expert mixture 0.0092 vs 0.2131.
- **unverified (external one-sided claims)**: 1.25 $\times$, 162 $\times$, 1.7 $\times$, 3.2 $\times$, +10.9 points, 26%, 30k concurrency, 381 chips, 2031 equivalent 1.4nm, compute 2–3 $\times$ estimate; "SeedFold" no evidence.

Appendix C Internal-review record (five-dimension self-assessment before submission, read-only)

- Novelty: medium-high (four-dimensional operationalization of the paradigm wall + falsifiable hypotheses for five paths + public correction of compute multiples).
- Evidence strength: internal weak (single seed), external medium (events multi-source verified, multiples not reproduced).
- Reproducibility: internal high (JSON rerunnable), external not applicable.
- Main gaps: ≥ 30 -seed recomputation; independent reproduction experiments of external papers; first-hand data on compute scope.

- Conclusion: **currently a discussable working paper, without submission status**; after completing F1–F4 it can target venues like TMLR / NeurIPS Datasets & Benchmarks / ICML workshops [CITATION NEEDED: venue selection needs separate deadline/partition checks].

Author's intended statement

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